



Operating System Capstone

Lab5 Exercise



Overview

- Ex 5-1 Thread Creation
- Ex 5-2 Frame buffer



task_struct

- A simple structure that stores process context in order to schedule a process.
- In lab5, more attributes may be needed in task_struct.

```
struct task_struct {  
    struct thread_struct {  
        unsigned long ra;  
        unsigned long sp;  
        unsigned long s[12];  
    } thread;  
    int pid;  
    unsigned long kernel_sp;  
    unsigned long user_sp;  
    unsigned long stack;  
    struct task_struct* next;  
};
```



Ex 5-1 : TODO

- Complete switch_to function in Start.S
- Implement schedule() function



switch_to(prev, next)

1. Save context into prev->thread_struct (ra, sp, s0-s11)
2. Restore context from next->thread_struct
3. Change tp register
4. Execute the second thread



schedule()

Find the next ready process from queue, perform context switch



Expected result

```
Starting kernel ...  
Process ID: 0x0000000000000003 0x0000000000000000  
Process ID: 0x0000000000000002 0x0000000000000000  
Process ID: 0x0000000000000001 0x0000000000000000  
Process ID: 0x0000000000000003 0x0000000000000001  
Process ID: 0x0000000000000002 0x0000000000000001  
Process ID: 0x0000000000000001 0x0000000000000001  
Process ID: 0x0000000000000003 0x0000000000000002  
Process ID: 0x0000000000000002 0x0000000000000002  
Process ID: 0x0000000000000001 0x0000000000000002  
Process ID: 0x0000000000000003 0x0000000000000003  
Process ID: 0x0000000000000002 0x0000000000000003  
Process ID: 0x0000000000000001 0x0000000000000003  
Process ID: 0x0000000000000003 0x0000000000000004  
Process ID: 0x0000000000000002 0x0000000000000004  
Process ID: 0x0000000000000001 0x0000000000000004
```



Ex 5-2 : TODO

- Implement usleep() function in main.c
- Enable playing the video from the frame buffer (On QEMU)
- just do it !!!!!!!!!!!!!!!!!!!!!

